

## Backtesting and EMA Strategy

### What is Backtesting?

Backtesting is a process used in trading and investing to test how a trading strategy would have performed in the past using historical market data. Instead of risking real money, the strategy is simulated on past prices to see whether it would have been profitable or not.

The key idea is:

- If a strategy performs well on past data, it **may** perform well in the future.
- Backtesting helps traders **evaluate profitability, risk, and consistency** of a strategy before applying it in live markets.

Commonly evaluated metrics in backtesting:

- **Total Return** – how much profit/loss the strategy made.
- **Sharpe Ratio** – risk-adjusted return.
- **Win Rate** – percentage of profitable trades.
- **Drawdown** – the largest drop from a peak to a low (risk measure).

### Backtesting with Exponential Moving Averages (EMA)

An **Exponential Moving Average (EMA)** is a type of moving average that gives **more weight to recent prices**, making it more responsive to market changes compared to a Simple Moving Average (SMA).

In this strategy:

- We use **two EMAs** – a short-term EMA (20 periods) and a long-term EMA (50 periods).
- **Buy Signal:** When the short EMA crosses **above** the long EMA → market trend turning bullish.
- **Sell Signal:** When the short EMA crosses **below** the long EMA → market trend turning bearish.

This is called a **crossover strategy**, widely used to capture medium-term market trends.

## PYTHON CODE AND OUTPUT FOR BACKTESTING

```

from backtesting import Backtest, Strategy
from backtesting.lib import crossover
from backtesting.test import GOOG  #Google Stock Data
import pandas as pd

def EMA(series, period):
    return pd.Series(series).ewm(span=period, adjust=False).mean()

class EmaCross(Strategy):
    n1 = 20  # Short-term EMA
    n2 = 50  # Long-term EMA

    def init(self):
        close = self.data.Close
        self.ema1 = self.I(EMA, close, self.n1)  # Short EMA
        self.ema2 = self.I(EMA, close, self.n2)  # Long EMA

    def next(self):
        # Buy when short EMA crosses above long EMA
        if crossover(self.ema1, self.ema2):
            self.position.close()
            self.buy()
        # Sell when short EMA crosses below long EMA
        elif crossover(self.ema2, self.ema1):
            self.position.close()
            self.sell()

bt = Backtest(GOOG, EmaCross,
              cash=10000, commission=.002,
              exclusive_orders=True)

output = bt.run()
print(output)
bt.plot()

```

```

Start                2004-08-19 00:00:00
End                  2013-03-01 00:00:00
Duration              3116 days 00:00:00
Exposure Time [%]    91.06145
Equity Final [$]     14649.35996
Equity Peak [$]      27874.16682
Commissions [$]      2131.90204
Return [%]           46.4936
Buy & Hold Return [%] 703.45824
Return (Ann.) [%]    4.58119
Volatility (Ann.) [%] 29.19315
CAGR [%]             3.13599
Sharpe Ratio         0.15693
Sortino Ratio        0.23584
Calmar Ratio         0.08201
Alpha [%]            14.42932
Beta                 0.04558
Max. Drawdown [%]    -55.8585
Avg. Drawdown [%]    -6.5942
Max. Drawdown Duration 1152 days 00:00:00
Avg. Drawdown Duration 79 days 00:00:00
# Trades              33
Win Rate [%]         30.30303
Best Trade [%]       91.74694
Worst Trade [%]      -17.9203
Avg. Trade [%]       0.79261
Max. Trade Duration  309 days 00:00:00
Avg. Trade Duration  86 days 00:00:00
Profit Factor        1.47785
Expectancy [%]       2.5128
SQN                  0.17876
Kelly Criterion       0.02647
_strategy            EmaCross
_equity_curve        ...
_trades              Size EntryB...
dtype: object

```